Interference

1]Interference in thin films: -

- In transmitted system

Constructive interference- $2\mu t \cos r = n\lambda$

Destructive interference- $2\mu t \cos r = (2n-1)\lambda/2$

- In reflected system

Constructive interference- $2\mu t \cos r = (2n-1)\lambda/2$
Destructive interference- $2\mu t \cos r = n\lambda$

Where,

- $\mu \rightarrow$ refractive index of the film
- $t \rightarrow$ thickness of the film
- $r \rightarrow$ angle of refraction
- $n \rightarrow$ order of interference
- $\lambda \rightarrow$ wavelength of the light

2]Wedge shape thin films: -

$$\beta = \frac{\lambda}{2\mu\theta}$$

Where,

- $\beta \rightarrow$ Fringe width

- μ → Permeability
- θ →Angle of wedge

$$\beta = \frac{\lambda}{2\theta} \text{ (for air film)}$$

Where,

- β→ Fringe width
- μ → Permeability
- θ →Angle of wedge

$$\tan\alpha = \frac{t}{l}$$

Where,

- t → Thickness
- l → Length

3]Newtons rings: -

$${D_n}^2 = 4Rn\lambda$$

Where,

- D_n →Diameter of n^{th} ring
- R → Radius of curvature
- n→ Order
- λ → Wavelength of light

$$D_{n+p}{}^2 = 4(n+p)R\lambda$$

Where,

- D_{n+p} = Diameter of $(n+p)^{th}$ rings
- R → Radius of curvature
- n→ Order
- λ → Wavelength of light

$$\lambda = \frac{D_{n+m}{}^2 - D_n{}^2}{4mR}$$

Where,

- D_n →Diameter of n^{th} ring
- R → Radius of curvature
- n→ Order
- λ → Wavelength of light

4]Anti and highly reflecting films: -

- For antireflecting films

$$t_{min}=\frac{\lambda}{4\mu_f}$$

- For highly reflecting films

$$t_{min}=\frac{\lambda}{2\mu_f}$$

Where,

- t_{min}→minimum thickness required
- λ→ wavelength of light used
- μ_f→ refractive index of film